

TwinMoE: Conserved Expert Models and State-Aware Computation for Battery Digital Twins

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Code and training: <https://github.com/fatasia/DeepMonkey-Studio/tree/main/research/battery-twin>

Abstract

Battery digital twins face two distinct errors: uncertainty about the physical state and approximation of its future dynamics. Increasing solver fidelity addresses the latter without necessarily resolving the former. We develop TwinMoE around this distinction. A modal diffusion solution supplies the particle inventory; tree and neural experts predict spatial corrections constrained to preserve it. A source-selected hierarchy combines an interpolation specialist with a constitutive neural expert. On 64 newly generated numerical controls, the hierarchy reduces concentration error by 30.3% and 40.8% relative to a single forest trained on the same data union, in near-domain and extended-control strata respectively. Relative to the reduced physical model, the reductions are 73.2% and 23.0%. For partially observed states, a weighted set of 27 health–diffusivity combinations reduces next-window voltage error by 24.0% relative to the best prefix fit. Sharing one neural correction across these states reduces measured CPU propagation cost by 5.29 times with a 0.027 mV mean change in prediction. Finally, a frozen one-query policy uses 12 rather than 66 SPMe evaluations in 12 charging decisions, reducing replay-accounted cost by 73.4%; both policies incur zero observed voltage violations, while unnecessary abstentions decrease from six to three. Boundary-first querying matches the selected policy’s outcomes. Numerical field references, measured-data subsystem evaluations and controlled aging interventions establish complementary evidence for conserved expert specialization and economical state-aware prediction.

1 Introduction

A battery twin is useful when it can compare actions that have not yet been executed. That requires more than reconstructing a measured voltage trace. The twin must propagate a plausible internal state under a proposed current, estimate where its approximation is inadequate, and decide whether further computation could change the preferred action.

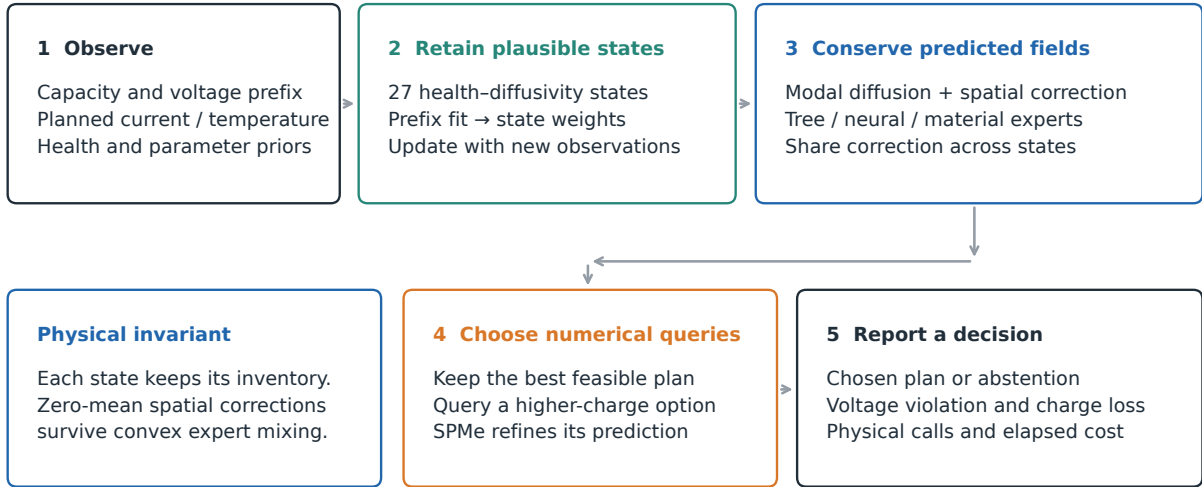
These requirements expose a distinction between information and computation. Several parameter–state combinations can explain a short voltage prefix. A more accurate solver can predict each combination more precisely, but it supplies no new observation to decide which combination describes the battery. Conversely, keeping many plausible states is expensive if each requires a separate neural field evaluation or electrochemical solve.

TwinMoE separates these problems through a common physical interface. Each state retains its own current-driven particle inventory. Heterogeneous experts correct the spatial distribution around

that inventory, while observations determine state compatibility. Numerical calls are then allocated among control plans that could improve the current decision. This architecture accommodates physical models, conventional machine learning and neural operators without requiring one model family to dominate every regime.

The study makes three contributions. First, an inventory-preserving correction interface supports a source-selected tree–neural hierarchy; a matched-data forest and new control draws test whether specialization adds value. Second, a shared-correction construction makes a finite state ensemble inexpensive while preserving each member’s physical inventory. Third, a charging study evaluates selective numerical calls against an early-stopping physical solver, reporting violations, abstentions, delivered-charge shortfall and computation together. The contribution lies in these constructions and their tested interactions, rather than in the presence of PINN, PINO or MoE components alone.

TwinMoE | information, approximation, and decisions



Solver output refines a prediction; only a new observation changes evidence about the physical state.

Figure 1: TwinMoE separates observation-driven state updates from approximation and decision computation. The final field hierarchy and the charging experiment use separately frozen expert configurations; charging results use the shared original neural correction, not a retrospective replacement by the final hierarchy.

2 Related Work

Electrochemical models and identification provide the physical basis. PyBaMM [1] implements the single-particle model (SPM), its electrolyte extension (SPMe), and the Doyle–Fuller–Newman model (DFN). Chen2020 [2] supplies the primary parameterization. Identifiability analysis of the SPM [3] shows that recovery of physical parameters depends on model structure and excitation; a low voltage residual alone is insufficient evidence of a uniquely recovered state.

Physics-informed neural operators combine solution data with physical constraints [4]. We investigate both spectral and local decoders, retaining the latter when source validation favors it.

The selected field expert is a physics-structured, nonspectral operator: an analytical diffusion prior, a causal diffusion-time memory and a learned spatial correction. A soft diffusion-residual recurrent baseline tests a related constraint mechanism under the same field task. This control does not stand for every PINN formulation.

Battery-specific expert mixtures already exist, including PIMOE for degradation trajectories [5]. BatteryMFormer [6] and PINN4SOH [7] address degradation forecasting and health estimation. These tasks differ from predicting local concentration under a proposed current. Our measured-data evaluations keep those targets separate. More generally, multifidelity methods [8] and decision-focused surrogate modelling [9] motivate allocating computation according to downstream use. Here the specific question is how to combine a conserved spatial approximation with uncertain physical states and a finite set of charging alternatives.

3 Methods

3.1 Physical state and available information

A request contains the observed voltage prefix, available capacity history, a scheduled current and temperature, initial particle concentrations, and effective diffusion and flux coefficients. Scheduled future current is a known control input; future voltage is unavailable during selection. Current is normalized by nameplate capacity. Health modifies effective inventory once, avoiding a second scaling of the same current.

The health branch uses the first 100 observed cycles and anchors its three-seed forecast at the last observed health. In controlled aging experiments, cached predicted and reference capacities determine estimated and reference active fractions. This uniform active-material scaling defines the numerical intervention. Rest-voltage samples estimate initial SOC using the parameterization’s OCP map. An effective polarization coefficient is fitted only on the available prefix. It is an input–output voltage correction, not an identified charge-transfer resistance.

3.2 Conserved expert corrections

The physical base solves finite-volume spherical diffusion. A volume-weighted symmetric modal decomposition integrates piecewise-linear current analytically, with the zero eigenvalue treated by its limit. The solution is repeated over electrode position. For each electrode species, let the spatial averaging operator combine uniform electrode-position weights with normalized radial shell volumes:

$$\langle z \rangle_w = \frac{1}{N_x} \sum_{i=1}^{N_x} \sum_{j=1}^{N_r} w_j z_{ij}, \quad \sum_j w_j = 1.$$

An expert predicts a residual R . The neural residual is anchored by subtracting its initial output; the tree residual sets its initial slice to zero because the tree is fitted on forecast slices. Both are centered and bounded:

$$q_0(t) = q(t) - \langle q(t) \rangle_w, \quad \hat{c}(t) = b(t) + \alpha(t)q_0(t), \quad 0 \leq \alpha(t) \leq 1.$$

The scale is common across electrode position and radius for each species and time. It is the largest permitted scale, capped at one, that keeps concentrations in $[0.001, 0.999]$ when the physical base is admissible. The neural branch also scales its correction by the maximum observed absolute current, capped at one.

Inventory property. The weighted mean of each correction is zero, so the predicted inventory equals that of b . For nonnegative expert weights summing to one, a mixture of corrections around the same base preserves this equality:

$$\left\langle \sum_e \pi_e(t) \hat{c}_e(t) \right\rangle_w = \langle b(t) \rangle_w, \quad \pi_e(t) \geq 0, \quad \sum_e \pi_e(t) = 1.$$

This property holds at each time even when the weights vary. It guarantees the integral constraint, not satisfaction of every local DFN equation. Common scaling matters: independent clipping at individual spatial points would generally alter the inventory.

The local neural expert uses a 16-channel diffusion-time memory conditioned on supplied diffusivities, spatial coordinates and the physical base. Two width-128 pointwise residual blocks with group normalization and GELU predict a bounded correction. The original expert has 36,674 parameters per seed. The constitutive extension adds four descriptors: log diffusivity ratios at concentrations 0.25 and 0.75 relative to 0.5 for the two electrodes. The Chen2020 ratios are zero; the descriptors distinguish the trained material domains.

3.3 Expert specialization

The tree expert predicts a residual from causal control tokens, accumulated charge, running maximum current and the radial physical base. It uses 96 ExtraTrees estimators with leaf size selected on source validation. A conserved output projection makes it compatible with the neural expert.

The hierarchy assigns the tree weight from distance to the source parameter box. The selected coordinates are the logarithms of the two dimensionless diffusivities and the two dimensionless flux coefficients:

$$d(z) = \left[\frac{1}{4} \sum_{j=1}^4 \left(\frac{\max(\ell_j - z_j, 0, z_j - u_j)}{s_j} \right)^2 \right]^{1/2}, \quad \pi_{\text{tree}}(z) = a \exp[-d(z)/\tau].$$

Bounds and scales come from training data, with a scale floor of 0.001. Source validation selects $a = 1$ and $\tau = 0.1$, with the constitutive neural model outside the box. Candidate choices also include ten-feature gates, other fixed outside experts and less abrupt decay. Original validation cells and four reserved Chen2020 material-validation controls receive equal total weight. The selected gate exploits a narrow original duration domain; it is an interpretable domain rule rather than a learned probability of model correctness.

3.4 State compatibility and shared correction

We retain 27 combinations: estimated health multiplied by 0.85, 1 or 1.15, and each electrode diffusivity multiplied by 0.5, 1 or 2. Health is clipped to $[0.5, 1.1]$. Each combination fits its polarization correction to the available prefix. If its prefix MAE is e , its normalized compatibility weight is:

$$\rho_k = \frac{\exp[-(e_k - \min_j e_j)/\eta]}{\sum_j \exp[-(e_j - \min_l e_l)/\eta]}, \quad \eta = 0.002 \text{ V}.$$

These are finite-model compatibility weights, not a calibrated posterior over all possible battery states. As observations arrive, the prefix fit and weights are recomputed. Forecasts use their weighted mean.

To avoid evaluating a neural field independently for every member, we calculate one nominal correction, remove its weighted mean, and apply it to each member’s own diffusion solution:

$$\hat{c}_k(t) = b_k(t) + \alpha_k(t)q_{\text{nominal},0}(t).$$

Member-specific common scaling preserves admissibility and inventory. This assumes the local discrepancy varies less strongly than the physical base across the selected parameter grid. A source-validation comparison against full per-member neural inference tests that approximation before adoption.

3.5 Decision-dependent numerical queries

The charging benchmark maximizes delivered charge among six plans with a common observation prefix and a peak-voltage threshold of 4.05 V. The threshold is an experimental decision criterion. Each expert has an empirical development margin selected by the same calibration rule. The weighted neural prediction uses 10 mV; SPMe uses 30 mV. Queries refine model predictions while leaving observation-based state weights unchanged.

At each step, the algorithm retains the highest-charge currently feasible plan. It considers unqueried plans with greater charge. The selected utility score multiplies potential charge gain by an exponentially decreasing function of distance above the voltage threshold, with state spread as scale and a 5 mV floor. A physical call replaces that plan’s estimated peak with the margin-adjusted SPMe result. An unresolved solve cannot certify feasibility. The source-selected budget is one query per group.

We compare this rule with boundary-first and charge-first queries, no queries, and SPMe early stopping. The latter evaluates plans by descending charge and stops at its first predicted feasible plan. Evaluation uses actual estimated-state SPMe outputs. Reference-state DFN outcomes determine violations and charge shortfall; they are never supplied to the selector.

4 Experimental Design

4.1 Data, domains and confirmation

Measured inputs originate from BatteryLife [10], the SNL cycling study [11], and the official PINN4SOH release [7]. Numerical internal fields are generated with PyBaMM 26.7.1.0 and sampled at 18 radial and eight electrode-position centers for each species. Native solver cell centers, rather than mesh endpoints, define the interpolation coordinates.

Evaluation	Population	Role
Original field data	14 training, 2 validation and 5 test source cells	Previously examined field benchmark
Material extension	12 training and 4 validation controls per material	Chen2020 and Ecker2015 specialist training
Final field confirmation	32 near + 32 extended numerical controls	Fresh draws after hierarchy and matched forest were frozen
Continuous confirmation	48 mixed-control numerical controls, 300–1,800 s	Independent duration distribution after gate freeze
Sequential state update	48 plans, 3 forecast windows, 5 health-source cells	Reused cohort; paired state comparison
Final charging study	12 groups, 6 plans each, 5 health-source cells	New plans after query policy selection
Measured voltage transfer	77 HUST + 64 NA-ion + 140 ZN-coin cells	Separate observation-prediction subsystem

The original requested source and test windows yield 66/96 and 20/30 complete numerical trajectories respectively. Failed and terminated trajectories remain recorded; those field metrics condition on completion. The final field confirmation completes all 64 requests. Near-domain controls last 300 s, use constant signed current of 0.03–0.55 C after an initial rest, and temperatures of 15, 25 or 35 degrees C. Extended controls last 600–1,800 s, use random knots, periodic reversals or ramps of 0.2–0.8 C amplitude, and temperatures from 10–40 degrees C. Initial SOC ranges from 0.2–0.7. Inputs include the known physical state and parameters.

An earlier 64-control diagnostic exposed failure of the original neural expert on extended controls. The final hierarchy was subsequently selected using source validation, followed by a second independent draw with seed 20270411 and frozen model-selection hashes. The final draw confirms the resulting method within the same two designed strata; it is not an untouched experimental battery cohort.

For material confirmation, eight further controls per trained material are generated after expert selection. This tests new controls within two trained parameter families. For charging, six plans per group target SPM peak levels of 4.03, 4.05 and 4.07 V before future DFN solves. All 72 final reference and SPM outcomes are known. Voltage-cutoff outcomes are counted as infeasible; numerical failures are unresolved rather than silently discarded.

4.2 Baselines and training

Original neural and recurrent comparisons share source-cell membership, a cell-balanced suffix field loss plus half the outer-shell MAE, and three seeds (2026, 3407, 7181). AdamW uses learning rate 0.001, up to 180 epochs, patience 25 and gradient clipping at one. Field loss excludes the first ten of 40 time samples. The recurrent controls use a width-64 GRU and a width-128 spatial decoder, with direct prediction, physical-prior residual prediction, or an additional soft physics penalty.

The soft penalty measures the exact finite-volume semigroup defect of the electrode-averaged correction plus inventory error; the known forcing cancels against SPM. Source validation selects weight 0.01 from 0.01, 0.1 and 1. This is a reduced solid-diffusion constraint, not the complete DFN PDE. Fourier, unconstrained and mean-only-prior ablations use the original field task.

Material neural models warm-start from the original weights and train for up to 160 epochs at learning rate 0.0003. A matched-data forest receives the union of original and material training data, the same constitutive descriptors and the same Chen2020 analytical prior. Original and material populations receive equal training and validation weight. Its 96-tree leaf-size search uses

2, 8 and 24. Raw and hard-conserved outputs are both evaluated. Thus, the final hierarchy cannot attribute its advantage merely to access to the broader training-data union.

4.3 Metrics, uncertainty and cost

Concentration is normalized by maximum concentration; error remains in that dimensionless unit. Voltage is reported in mV. Original field and health studies average within source cell before averaging across cells. Final control confirmation averages trajectories separately in each stratum. Ten thousand paired bootstrap resamples use controls for field confirmation and source cells for health/state comparisons. Intervals are exploratory and unadjusted for multiple comparisons. For the state-set forecast, we also report a split-conformal diagnostic: development-cell residuals calibrate a cutoff-specific absolute-error radius, and untouched test cells evaluate coverage. This is a calibration check for the finite state ensemble, not a posterior probability claim.

Charging shortfall is oracle-feasible charge minus safely delivered selected charge; abstention or violation receives zero delivered charge. We report group means and cell-paired uncertainty separately. Twelve plan groups reuse five health-source cells, so their outcomes do not establish a population failure probability.

CPU inference uses four Torch threads and preloaded weights. Forest inference is timed with one and four workers; one is faster for these small requests. Direct baselines exclude unused physical-prior work. Charging costs sum measured state inference and selected physical calls, including work used to make the selection. They are replay-accounted costs, separate from cold process-launch latency.

5 Results

5.1 A common physical structure, different useful experts

On the original held-out field controls, the conserved local decoder has MAE 0.000204 versus 0.001829 for modal SPM, an 88.9% reduction. Its long-plan MAE is 0.001468 versus 0.003293, a 55.4% reduction. Removing the radial prior increases long-plan error to 0.006092. The physical base carries essential spatial information.

Strong conventional controls change the interpretation of these gains:

Original benchmark method	Cell MAE $\times 1000$	Long-plan MAE $\times 1000$
Modal SPM	1.829	3.293
Direct GRU	6.214	33.723
GRU + physical prior	1.070	2.501
GRU + soft diffusion constraint	1.070	2.520
Direct ExtraTrees	0.097	35.028
ExtraTrees + physical prior	0.048	2.655
Conserved local neural	0.204	1.468

The prior-assisted forest is the strongest original-domain interpolator. The neural model transfers better to the original long-plan set, but this ranking does not hold for every new control family. In the first 64-control diagnostic, its extended-control MAE rises to 0.001815 while SPM attains 0.000756. Reusing the broader constitutive expert reduces that error to 0.000496. This failure and recovery motivate specialization.

The local decoder also does not dominate every architectural ablation. A smoothed Fourier variant has slightly lower original cell MAE (0.000202), while its long-plan error is higher (0.001622).

An unconstrained local decoder attains 0.001428 on long plans, with inventory drift up to 0.00231; the conserved version remains at approximately 0.0000001. The structural gain is predictable inventory, not a promise of minimum pointwise error.

5.2 Frozen hierarchy on independent control draws

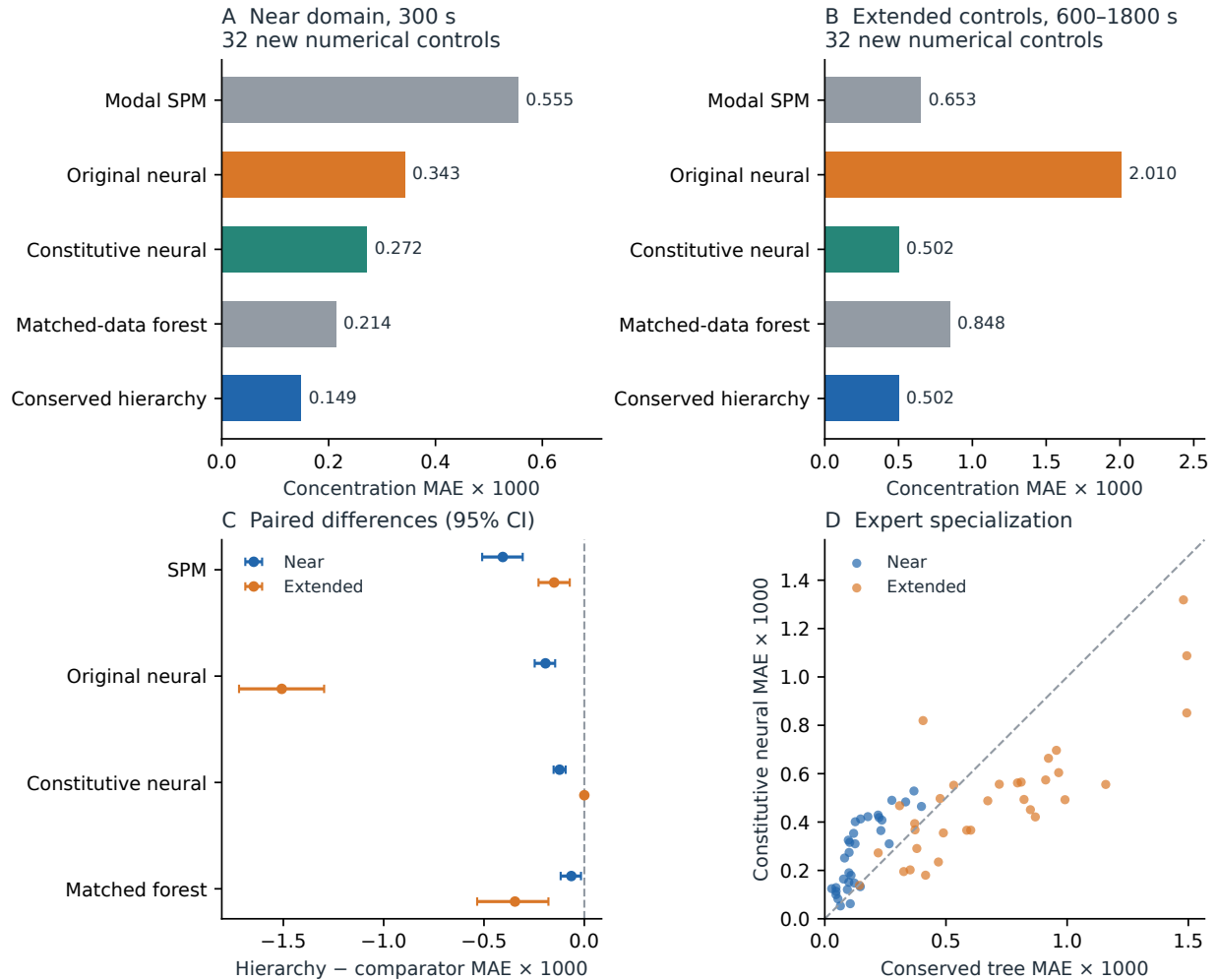


Figure 2: Final field confirmation. A–B: stratum means on 32 controls each. C: paired mean error differences with exploratory 95% control-bootstrap intervals. D: errors of the two constituent experts on each control. The hierarchy selects the conserved tree throughout the near stratum and the constitutive neural expert throughout the extended stratum.

The final hierarchy reaches MAE 0.000149 near-domain and 0.000502 on extended controls. The matched-data forest yields 0.000214 and 0.000848: relative reductions of 30.3% and 40.8%. Paired hierarchy-minus-forest intervals are $[-0.000118, -0.0000167]$ and $[-0.000534, -0.000179]$. Its conserved forest variant yields 0.000221 and 0.000848.

Relative to SPM, the hierarchy reduces error by 73.2% and 23.0%. Near-domain, it improves on the constitutive neural expert by 45.3%; in the extended stratum, it improves on the conserved tree by 28.1%. It equals the selected constituent within each stratum. The benefit is selecting

complementary fixed experts across the two domains, not a gain beyond an oracle constituent choice.

To test whether this result was only a two-bin duration rule, we froze the gate and drew 48 mixed-control cases with durations uniformly spanning 300–1,800 s, four control shapes, temperatures of 10–40 degrees C and SOC of 0.18–0.74. Every solve completed. The frozen route gives MAE 0.000361 (median 0.000313; 95th percentile 0.000827), versus 0.000553 for SPM and 0.000690 for the matched forest. This is a known-state numerical confirmation, not a measured-cell generalization claim.

The same holdout is an adversarial router check. All 48 continuous cases receive zero tree weight; the 64-case two-stratum confirmation contains 32 tree-routed and 32 constitutive-routed cases, with no strictly mixed weights. A diagnostic fixed blend is therefore reported separately: tree weights of 0.25 and 0.50 give final-confirmation MAEs of 0.000349 and 0.000348, while the selected source gate gives 0.000326 on the same controls; on the continuous holdout the 0.25 blend gives 0.000357 versus 0.000361 for the constitutive branch. These blends were not selected from holdout labels. The result supports source-stratified expert specialization, but does not establish a universally adaptive soft router.

Single-request execution agrees with the frozen batch evaluation to a maximum absolute concentration difference of 0.00000012 across all 64 controls. Its maximum inventory discrepancy is of the same numerical order. Warm CPU requests on four prechosen controls take 6.6–18.7 ms. The matched forest with one worker takes 4.6–4.8 ms; specialization improves accuracy at a higher inference cost. The original neural-versus-warm-DFN benchmark measures 11.02/17.22 ms against 524.60/835.03 ms on its two timing sets.

Reference-resolution sensitivity uses the first four controls in each stratum, selected by index. With 60 nodes per domain and solver tolerances of 0.00000001, near-domain MAEs are 0.000183 for the hierarchy, 0.000287 for the matched forest and 0.000635 for SPM; extended-control MAEs are 0.000829, 0.001611 and 0.000995. Changes from 40 to 60 nodes average 0.00000546 and 0.00000761 respectively. The mean ranking persists on this subset, although the hierarchy does not win every individual control.

As a solver-sensitivity check, eight prespecified continuous controls were re-solved with PyBaMM’s CasadiSolver using the same Chen2020 electrothermal DFN parameters. All eight completed. The maximum voltage difference from the IDAKLU reference was 0.157 mV; field MAE between solvers was 0.00000990 and the maximum field difference was 0.000123. This checks numerical solver sensitivity, not independent chemistry or experimental truth.

For the separately selected material experts, fresh Chen2020 controls yield MAE 0.000406 versus SPM’s 0.000910; Ecker2015 yields 0.000242 versus 0.000280. The Ecker expert corrects an actual SPM prior, so its value is an accuracy refinement rather than elimination of that solve. Eight-control paired intervals favor the selected expert in each trained material domain.

5.3 State uncertainty is worth propagating

In the health-to-field experiment, predicted-health neural fields reduce MAE from 0.002892 for same-health SPM to 0.001456. Holding the last observed capacity instead gives 0.001705; the health forecast improves this stronger control by 14.6%, with a five-cell paired interval of [-0.000464, -0.0000352]. Estimated-state DFN still has MAE 0.000496, despite sharing the reference solver family.

Jointly fitting more parameters does not reliably fix this uncertainty. In an eight-case inverse study, fitting SOC, health, two diffusivities and polarization reduces prefix voltage error from 7.47 to 5.14 mV, yet increases future error from 17.76 to 25.47 mV relative to fitting only health and

polarization.

Keeping alternatives is more effective in the sequential study. Across 144 next-window forecasts, the weighted 27-state predictor has mean voltage MAE 5.308 mV versus 6.981 mV for the single best prefix fit, a 24.0% reduction. The five-cell paired difference is -1.673 mV, with interval [-2.340,-0.961]. The shared-correction version has MAE 5.299 mV and differs from full-member forecasts by only 0.027 mV on average. Paired CPU timings on eight requests decrease from 233.5 to 44.1 ms, a 5.29-fold reduction.

Split-conformal calibration makes the uncertainty claim more specific. Calibrating on development cells and evaluating the five-cell test cohort gives 98.3% point coverage and 93.1% full-trajectory coverage at the nominal 90% level, with a mean interval width of 57.4 mV. The weighted center has 5.685 mV MAE. The wider finite-state envelope reaches 99.8–100% point coverage but spans 111–127 mV, so it is conservative rather than sharply calibrated.

5.4 Querying fewer alternatives

Final charging policy	Violations	Unnecessary abstentions	SPMe calls	Total cost (s)
Shared states, no query	0/12	5	0	2.39
One utility query per group	0/12	3	12	7.44
One boundary query per group	0/12	3	12	7.50
One charge-first query per group	0/12	5	12	7.55
SPMe with early stopping	0/12	6	66	27.99

The frozen one-query policy reduces physical calls by 81.8% and replay-accounted cost by 73.4% relative to SPMc early stopping. Mean charge shortfall decreases from 0.083665 to 0.046855 C-rate hours. Budgets of two or three queries per group produce the same final outcomes. Boundary-first matches utility-first, so the experiment supports selective querying without isolating a benefit of the utility formula over that simpler rule.

The cell-macro paired shortfall difference versus early stopping is -0.02945 C-rate hours, with interval [-0.07212,0]. Small-cohort decision evidence is therefore less conclusive than the field comparison. Zero observed violations under empirical margins is not a calibrated future-risk guarantee.

The budget curve shows where the gain comes from: zero, one, two and three utility-guided queries use 0, 12, 24 and 34 physical calls, with shortfalls 0.0747, 0.0469, 0.0469 and 0.0469 C-rate hours. Early-stopping SPMc uses 66 calls and has 0.0837 shortfall. These are 12 paired groups on five independent health-source cells; additional randomized plans increase control coverage, not the number of independent batteries.

5.5 Measured-data subsystems and unsuccessful extensions

The frozen raw-sampling voltage mixture attains 1.5205 mV on 77 HUST cells, versus 1.8684 mV for linear trend. All 77 cells improve; the paired interval is [-0.3595,-0.3358] mV. Current noise of 0.05 C reverses the advantage. On 28 Farasis large-step records, persistence is better (3.542 versus 8.508 mV).

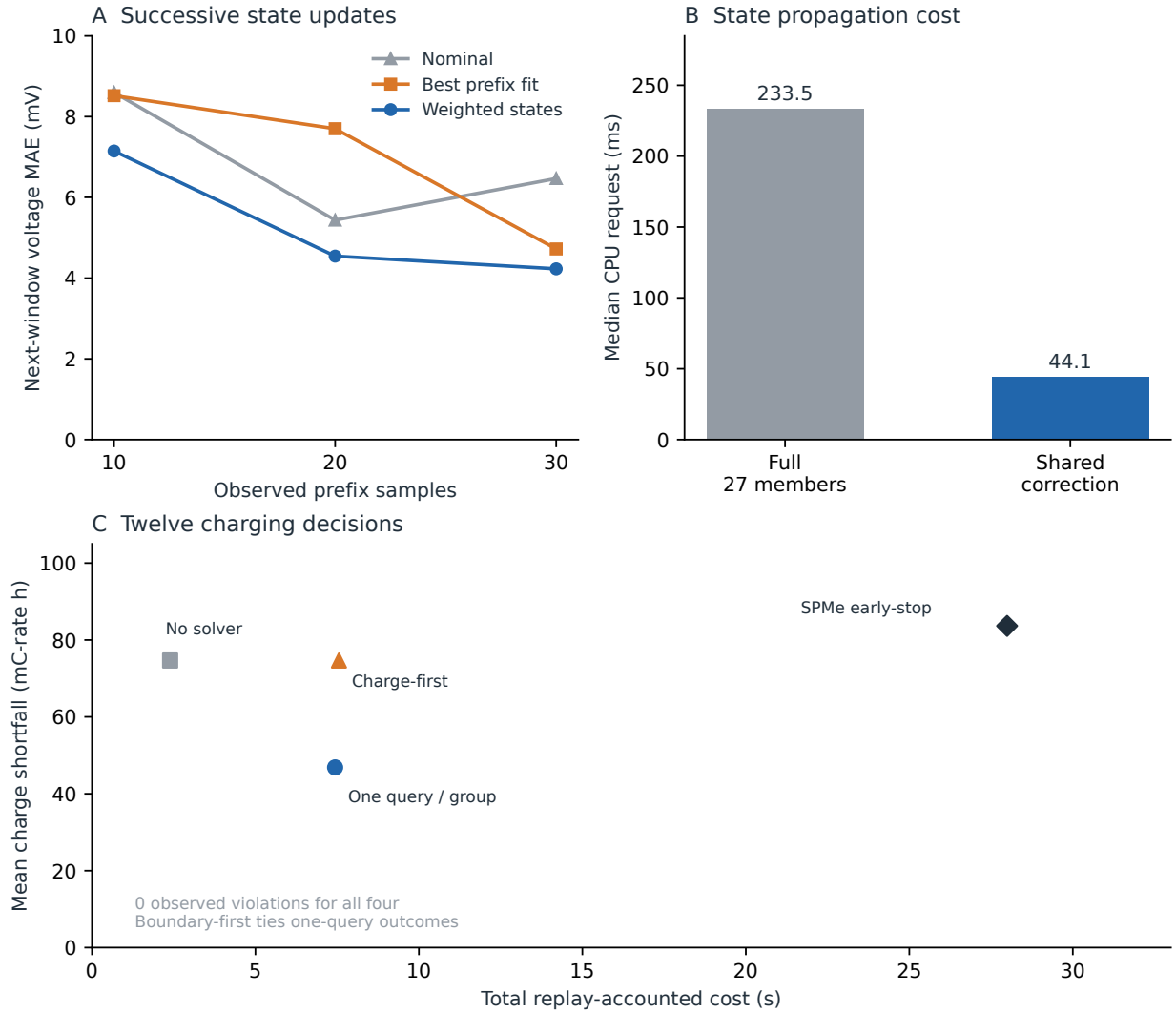


Figure 3: State and decision results. A: next-ten-sample voltage error after 10, 20 and 30 observed samples; 48 plans and five reused health-source cells. B: paired CPU propagation on the first eight requests. C: total replay-accounted cost and mean charge shortfall across 12 final groups; zero violations are observed for the shown policies. Boundary-first one-query selection matches the selected policy’s outcomes.

New BatteryLife source evaluations contain 578,756 samples from 64 NA-ion and 140 ZN-coin cells, including 1,737 predeclared steps above 0.1 C. The mixture improves step error over trend on both sources, but trend wins overall on NA-ion and persistence wins overall on ZN-coin. A source-selected online adapter fails to improve both sources and is not adopted.

A frozen paired-cell audit puts the measured subsystems on one statistical footing without pooling their targets. On 77 HUST cells, the native mixture improves over trend by 0.348 mV per cell (95% bootstrap interval [0.336, 0.360] mV; Wilcoxon $p < 0.001$). On NA-ion, trend remains better on all samples by 0.340 mV, while the mixture improves the predeclared current-step subset by 9.212 mV; on ZN-coin, the step-subset gains are 12.776 mV over trend and 12.608 mV over persistence. In the separate 77-cell HUST SOH comparison, PINN improves RMSE but not MAPE or MAE.

On the separate same-cycle health-estimation task, the current official PINN4SOH implementation achieves 1.4209% cell-macro MAPE against 1.4506% for a physical-feature ExtraTrees control using the same finite rows and training-only normalization. Their paired interval crosses zero. This reproduction does not compare our field errors with the published paper’s SOH percentages.

As an external product check, we replayed 20 vehicle-isolated BAIC EU500/CATL NCM charging traces (all records held out from training). The causal assimilation candidate uses one initial BMS SOC observation and no future SOC labels: mean SOC MAE is 6.944 percentage points versus 6.986 for calibrated Coulomb counting and 7.033 for SOCFormer. The neural correction improves 8/20 vehicles and is disabled by its stability gate on the remainder. All 20 traces are correctly routed out of the LFP PINO domain to the SPM fallback. These are BMS-referenced weak labels, so the result supports cross-domain synchronization and safe routing, not laboratory SOC accuracy.

Two additional frozen external checks separate source transfer from internal-field claims. Zero-shot Oxford evaluation on eight cells gives a cell-macro SOH MAPE of 2.708%; a McMaster LG HG2 leave-one-temperature-out laboratory check improves the capacity reference on four of six temperature folds, but uses one physical cell. Neither result supplies measured internal concentration fields or cell-independent decision validation.

Not every proposed mechanism helps. Adding PDE and domain signals reduces failure-detection AUC from 0.941 to 0.894; current amplitude alone attains 0.888. A finite state-plan refinement experiment requires a 0.265 V global numerical-error envelope. On development data its decision-priority strategy needs 662 physical queries to reach the full-grid result, versus 623 for an exact early-stop control. It is excluded from the selected system. These findings favor a compact implementation over additional machinery without measured benefit.

6 Discussion

The strongest result is a practical form of specialization under a shared physical invariant. Conventional trees are valuable near the training domain. A broader neural correction handles the tested time and control variation more effectively. Enforcing the same inventory makes their outputs composable and keeps model choice from changing the total lithium implied by the supplied state. This provides an explicit interpretation of the experts and gate.

State-aware computation adds a second, independent economy. The physical base captures most parameter variation, allowing a nominal correction to be reused without materially changing the tested voltage forecasts. The resulting ensemble avoids treating the minimum prefix residual as a uniquely determined state. Numerical queries then address approximations of decision-relevant alternatives. They cannot resolve state ambiguity without additional observation.

The evidence supports these mechanisms within specified domains. The continuous confirmation reduces dependence on the original two duration bins, but its router stress test shows that the frozen source gate is still effectively stratified rather than a universal soft router. All field results use known numerical states. The same five health-source cells recur in several decision studies. Material extension includes both families in training, while the internal-field reference remains one PyBaMM/DFN solver family; an eight-control CasadiSolver check bounds solver sensitivity but does not supply independent chemistry. The state grid is finite; split-conformal bands improve calibration, but the finite-state envelope remains wide. The simple boundary query ties the selected policy, while the broader state-plan bound scheme fails its development comparison. Independent chemistry, cross-solver physics, non-rest initialization and measured intervention outcomes would strengthen the general decision claim.

7 Reproducibility and Data Availability

Measured data sources are BatteryLife [10], SNL [11] and the official PINN4SOH repository [7]. Numerical references use PyBaMM [1] with Chen2020 [2] and its bundled Ecker2015 parameterization. Source/cell membership, normalization, failed solves, selection hashes, seeds, saved predictions and per-control errors accompany the local review package. Third-party data and weights retain their source licenses.

Learning uses Python 3.13.9 and PyTorch 2.12.0+cu130; numerical generation uses Python 3.12 and PyBaMM 26.7.1.0. Source selection, two-stage field confirmation, state updates, physical-call replay and runtime parity have separate executable entry points. Figure data are exported to CSV, and vector PDF/SVG versions accompany the manuscript. Local file-order checks document execution order; they are not external preregistration.

The code and training entry is <https://github.com/fatasia/DeepMonkey-Studio/tree/main/research/battery-twin>.

8 Conclusion

TwinMoE combines heterogeneous battery predictors through a conserved correction interface and separates state ambiguity from numerical approximation. Frozen-control confirmation supports complementary tree-neural specialization; sequential forecasts support inexpensive propagation of multiple plausible states. Selective physical calls reduce computation in the charging benchmark, with simpler boundary selection matching the chosen utility rule. These results identify where learning, physical simulation and expert selection each contribute to a useful battery twin.

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